

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\emmaj\OneDrive\Desktop\hw2-emma`

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Problem 1 Write Up

Overall, this indicates an overall positive feedback cycle. As atmospheric CO₂ increases, the relationship between atmospheric CO₂ and ocean CO₂ concentration weakens as the ocean's capacity to store CO₂ weakens. And therefore, this leads to an overall increase in global mean temperature with no method of carbon sink.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.1 Write Up

We added two boxes, as there are two mixing, and therefore two discharges.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.2 Write Up

Problem 2.3

Determine if the system is in compliance with a regulatory limit of 2.3 kg/(1000m³). You can do this analytically or computationally.

Problem 2.3 Write Up

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “[Snowball Earth](#)”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1 - \alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m²/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;

- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.1 Write Up

Using the governing Euler's Method equation, the climate model equation was substituted in and rearranged until we ended up with the final equation.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```

deltt = 1
S = 1272
A = 221.2
B = -1.3
C = 51
Tmax = 200
T = range(-60,30)

function comp_alpha(T)
    if T <= -10
        return 0.5
    elseif T >= 10
        return 0.3
    else
        return 0.5 + (0.3-0.5)*((T+10)/20)
    end
end

function climate_model(S, A, B, C, T)
    alpha = comp_alpha(T)
    return (((S*(1-alpha))/4)- A + (B * T))/C
end

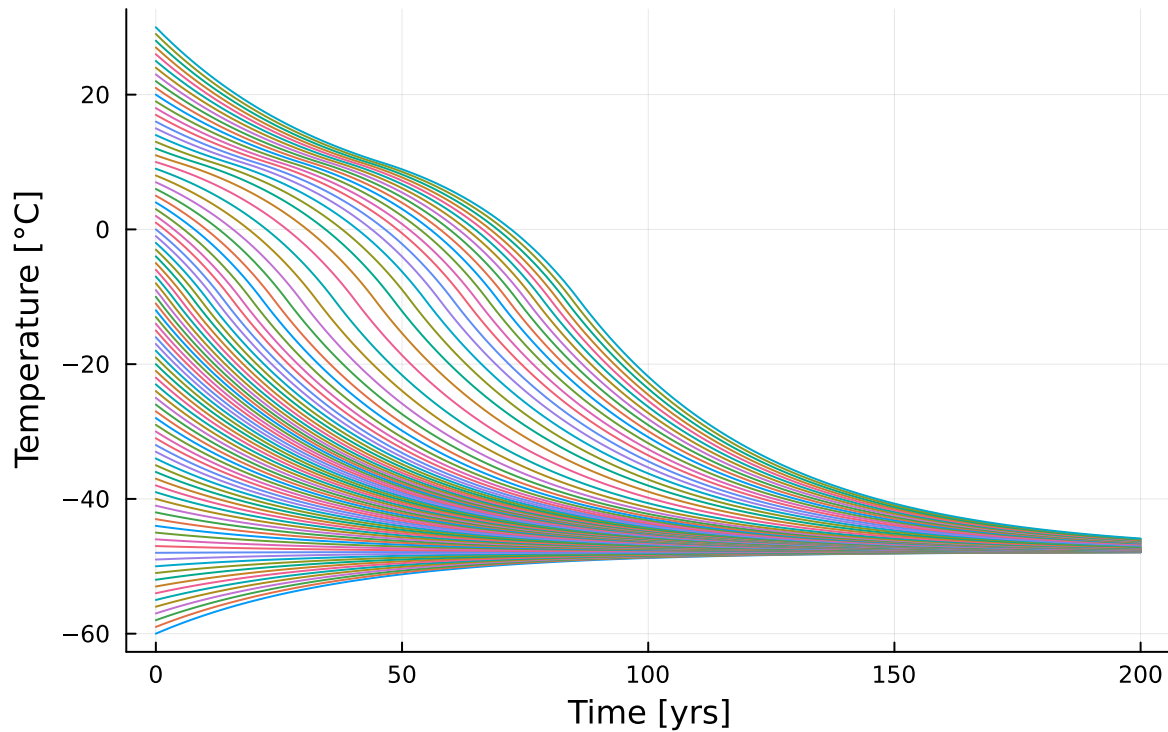
p = plot(xlabel="Time [yrs]", ylabel="Temperature [°C]", title="Temperature Trajectories")

for boo in T
    empty = zeros(Tmax + 1)
    empty[1] = boo
    for omnom in 1:Tmax
        empty[omnom+1] = empty[omnom] + deltt * climate_model(S, A, B, C, empty[omnom])
    end
    plot!(p, 0:Tmax, empty, label = "")
end

display(p)

```

Temperature Trajectories



Problem 3.2 Write Up

This code was largely derived from Professor Viveks presentation on Simulation Models, where the derived euler method for this problem that was solved in 3.1 was implemented in code form via “func climate_model()”, where respective constant from the problem were defined and substituted in. To account for the changing alpha, a function was created (comp_alpha()), where based on the T value the respective alpha value from the piecewise function would be computed. After running these function, they were plotted on the graph as temperature (in celsius) over time (in years).

According to this graph, one stable equilibrium was found at around a little under -50 degrees celsius (~ -47 degrees celcius).

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

Problem 3.3 Write Up

No, the increasing amount of solar radiation is not enough to explain the warming temperatures due to the Snowball effect. This calculation shows that the sun would have to emit way more energy than physically possible in such a short amount of time to justify the rapidly increasing temperature of the Earth.

References

List any external references consulted, including classmates.

The base functions to code the Euler Method as well as how to plot it were directly from Professor Vivek's code from the 9/15 Simulation Models presentation. This project was collaborated with Pranati Patnam.